Databases 2.0

14th November 2024

Marking Scheme

This marking scheme has been prepared as a **guide only** to markers. This is not a set of model answers, or the exclusive answers to the questions, and there will frequently be alternative responses which will provide a valid answer. Markers are advised that, unless a question specifies that an answer be provided in a particular form, then an answer that is correct (factually or in practical terms) **must** be given the available marks.

If there is doubt as to the correctness of an answer, the relevant NCC Education materials should be the first authority.

Throughout the marking, please credit any valid alternative point.

Answer ALL questions

Marks

Question 1

A university database tracks information about students, modules, professors and the departments to which they belong.
Each student is enrolled in multiple modules and each module is taught by one or more professors. Professors are assigned to specific departments and each department offers multiple modules.

a) Identify FOUR (4) entities in the university database system based on the scenario provided. **4**

Mark scheme:

Student, course, professor, department

1 mark for each, up to a max of 4 marks.
Any suitable answers that are related to the scenario should be accepted.

b) Identify an example of a primary key that could be used in one of the entities you've identified in a). Explain how a primary key can be used to uniquely identify records. **3**

Mark scheme:

In the student entity, the primary key could be student_id (1 mark) which would be a unique value assigned to each student (1 mark). The key ensures that each student is uniquely identified in the database (1 mark), preventing duplicate entries (1 mark).

1 mark for any point above, up to a maximum of 3 marks.

Any suitable answers that are related to the scenario should be accepted.

c) Identify an example of a foreign key that could be used in the given scenario. Explain how it would ensure referential integrity. **3**

Mark scheme:

course_id in a table that tracks student enrolments (1 mark). It links the enrolment entity to the course entity (1 mark). The course_id in the enrolment would be a foreign key referencing the course_id primary key in the course table (1 mark). This ensures that every course a student enrols in exists in the course table, maintaining referential integrity (1 mark).

1 mark for any point above, up to a maximum of 3 marks.

Any suitable answers that are related to the scenario should be accepted.

Total 10 Marks

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Marks

Question 2

A small online shopping platform uses a relational database to manage user accounts, transactions and product listings. Each day, the system handles several transactions and the platform must ensure that all operations run smoothly and consistently.

a) Identify each property in ACID. Provide ONE (1) practical example of how each property would help ensure smooth operations for the online shopping platform. **8**

Mark scheme:

Award 1 mark for identifying each property and 1 mark for each suitable example, up to a maximum of 8 marks.

Atomicity: if a customer makes a purchase, the system must deduct the correct amount from their account and mark the product as sold. If the system fails halfway, neither action should occur.

Consistency: if a customer's balance is deducted, the total sales revenue must be updated accordingly.

Isolation: if two customers try to buy the same product at the same time, isolation ensures that only one transaction will succeed in reserving the product.

Durability: once a customer makes a successful purchase, the transaction will still be recorded even if the system crashes immediately after.

b) Explain the purpose of the ROLLBACK command in SQL. Identify a situation where it would be used. **2**

Mark scheme:

The rollback command is used to undo all changes made during the current transaction (1 mark), restoring the database to its previous state (1 mark). It's useful if an error or unexpected event occurs during a transaction (1 mark), ensuring that incomplete or incorrect changes are not saved to the database (1 mark).

A situation could be if a customer's payment is processed, but an error occurs when updating the order record (1 mark). Rollback would ensure that the customer is not charged without placing an order (1 mark).

1 mark for the explanation, and 1 mark for an example.

Total 10 marks

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Marks

Question 3 **10**

Scenario:

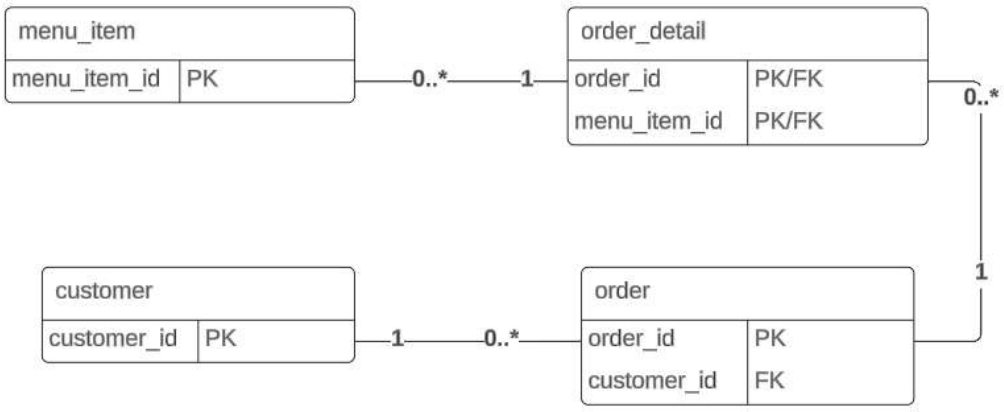
A restaurant wants to create a database to manage its customers, menu items and the orders placed by customers.

Customers can place multiple orders and each order can contain ONE (1) or more menu items. Menu items can be part of many orders.

Draw an Entity Relationship Diagram for the restaurant database that includes FOUR (4) entities: menu item, order detail, order and customer.

You need to identify relationships, cardinalities and label all the primary and foreign keys. Clearly state which entity the keys belong to.

Mark scheme:



1 mark for all relationships and 1 mark each for cardinalities. The notation types should not be the focus of this marking, up to a max of 4 marks.

1 mark for each correctly labelled PK and FK, up to a maximum of 6 marks. An order_details_id as PK should be accepted but must be along with order_id and menu_item_id as FK.

Total 10 marks

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Marks

Question 4

a) A smart parking system uses a database to manage parking spots, cars and parking reservations. **4**

Entities:

- Parking spot
- Car
- Reservation

Create a CRUD matrix to show the following transactions:

Transaction 1 – Add a new parking spot to the system

Transaction 2 – Create a new parking reservation for an existing car

Transaction 3 – Update car information for an existing reservation

Transaction 4 – Remove a parking spot that is no longer available

Mark scheme:

Award 1 mark for each correct row in a table, up to a maximum of 4 marks. 0.5 mark should be given for a partially correct row. Marks should be rounded up at the end of the question.

Transaction	***Parking spot***	***Car***	***Reservation***
T1	***C***		
T2		***R***	***C***
T3		***U***	***R***
T4	***D***		

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Marks

b) Explain data integrity. **1**

Mark scheme:

Data should be logically consistent and accurate. (1 mark)

Any other suitable answers should be accepted.

c) Identify **and** explain TWO (2) potential issues that could arise if data integrity is not enforced when removing a parking spot that is currently reserved or occupied. **4**

Mark scheme:

- ***Double Booking (1 mark): If a reserved or occupied parking spot is removed without ensuring data integrity, the system might not update the reservation status correctly. This could lead to the same spot being assigned to multiple users simultaneously (1 mark).***

- ***Loss of Reservation Data (1 mark): Removing a parking spot without proper data integrity checks can result in the accidental deletion of reservation records. This means that users who had valid reservations might lose their booking information (1 mark).***

1 mark for each suitable issue above and 1 mark for its explanation, up to a max of 4 marks.

Any other suitable potential issues should be accepted.

d) Identify ONE (1) database mechanism that can be used to maintain data integrity in the situation described in c). **1**

Mark scheme:

Foreign key constraint (1 mark): ON DELETE RESTRICT (1 mark).

1 mark for any one point above.

Total Marks 10

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Marks

Question 5

a) A local gym maintains a spreadsheet to track memberships and training sessions. The spreadsheet records each member's ID and full name, along with the names of the trainers and the dates for training sessions. **8**

Each member can have multiple training sessions, but they can only work with one trainer per session.

The current structure of the data does not conform to the First Normal Form. Convert the provided data into 1NF. Make sure you clearly identify keys.

Provided data:

member_id	member_name	training_sessions
201	Tom Johnson	“Strength training” with Luke “Kick boxing” with Daniel
202	Alice Smith	“Weightlifting” with Ben
203	Chris Lee	“Cardio” with Luke “Pilates” with Sarah “Core training” with Daniel

Mark scheme:

member_id(PK)	***first_name***	***last_name***
201	***Tom***	***Johnson***
202	***Alice***	***Smith***
203	***Chris***	***Lee***

session_id(PK)	***session_name***	***trainer_name***	***member_id (FK)***
1001	***Strength training***	***Luke***	***201***
1002	***Kick boxing***	***Daniel***	***201***
1003	***Weightlifting***	***Ben***	***202***
1004	***Cardio***	***Luke***	***203***
1005	***Pilates***	***Sarah***	***203***
1006	***Core training***	***Daniel***	***203***

1 mark for dividing the member_name column into two separate columns: first_name and last_name.

1 mark for each correctly identified table, up to a maximum of 2 marks.

1 mark for including member_id in the session table and 1 mark if it’s identified as an FK.

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1 mark for the presence of session_name and trainer_name in the session table.

1 mark for including member_id in the member table and 1 mark if it's identified as a PK.

b) The gym is considering switching from a spreadsheet to a database system for managing memberships and training sessions. Identify TWO (2) advantages of using a database system compared to the current spreadsheet. **2**

Mark scheme:

1 mark for each correct advantage, up to 2 marks. Suggested answers are not limited to:

- ***Data integrity***
- ***Data consistency / No duplication***
- ***Efficient data retrieval***
- ***Automatic report generation***
- ***Scalability / handle large amount of data***
- ***Concurrent access by multiple users***
- ***Backup / security controls***

Any other suitable answers should be accepted.

Total 10 marks

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Marks

Question 6

Review the scenario in Question 4. It is shown below for your convenience:

A smart parking system uses a database to manage parking spots, cars and parking reservations.

Entities:

- Parking spot
- Car
- Reservation

Write the appropriate SQL command to perform each of the following tasks to the parking_spot data provided below.

parking_spot

spot_id	location	charge_per_hour (£)	spot_type
1	AB1	2.00	Electric
2	BG2	1.50	Regular
3	EN8	2.99	Regular

a) Create the `parking_spot` table. **5**

Mark scheme:

CREATE TABLE parking_slot(

spot_id INT NOT NULL PRIMARY KEY,

location VARCHAR(10),

charge_per_hour DECIMAL,

spot_type VARCHAR(10)

);

1 mark for each correctly defined row in the SQL command, up to a maximum of 5 marks. Allow minor syntax errors (commas missing, etc)

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Marks

b) The `parking_spot` table needs to be updated to include a new column named `is_reserved` to track whether a spot is reserved or not. **2**

Mark scheme:

```
ALTER TABLE parking_spot
ADD is_reserved BOOLEAN;
```

1 mark for each correctly defined row in the SQL command, up to 2 marks.

c) The `charge_per_hour` for the parking slot with spot_id 3 needs to be updated to £3.50. **3**

Mark scheme:

```
UPDATE parking_spot
SET charge_per_hour = 3.5
WHERE spot_id = 3;
```

1 mark for each correctly defined row in the SQL command, up to 3 marks.

Total 10 marks

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Marks

Question 7

Review the scenario described in Question 4. The database contains the following tables.

parking_spot

spot_id(PK)	location	charge_per_hour (£)	spot_type
1	AB1	2	Electric
2	BG2	1.5	Regular
3	EN8	2.99	Regular

car

licence_plate(PK)	owner_name
XYZ1234	Chris Lee
ABD1234	Alice Smith
BDF1234	John Brown

reservation

reservation_id(PK)	start_time	end_time	licence_plate(FK)	spot_id(FK)
301	2024-12-23 09:00:00	2024-12-23 10:00:00	XYZ1234	1
302	2024-11-02 12:00:00	2024-11-02 14:10:00	ABD1234	2
303	2024-10-14 08:00:00	2024-10-14 11:30:00	BDF1234	2
304	2024-09-20 19:00:00	2024-09-20 22:00:00	ABD1234	3
305	2024-09-16 11:00:00	2024-09-16 13:00:00	BDF1234	1

a) Write an SQL command to display all the reservations that take place on parking spots with the spot_type 'Regular'. **3**

Mark scheme:

```
SELECT r.* FROM reservation r
JOIN parking_spot ps ON r.spot_id = ps.spot_id
WHERE ps.spot_type='Regular';
```

1 mark for each correctly defined row in the SQL command, up to a maximum of 3 marks.

b) Write an SQL command to display each car's licence_plate, owner_name and the location of the parking spot it was reserved in. **3**

Mark scheme:

SELECT c.licence_plate, c.owner_name, ps.location (1 mark)
FROM car c
JOIN reservation r on c.licence_plate = r.licence_plate (1 mark)
JOIN parking_spot ps ON r.spot_id = ps.spot_id; (1 mark)

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Marks

c) Write an SQL command to list each car's licence_plate, owner_name and the total number of reservations it has made. **4**

Mark scheme:

SELECT c.licence_plate, c.owner_name, COUNT(r.reservation_id) (1 mark) AS total_reservations (1 mark)
FROM car c
JOIN reservation r ON c.licence_plate = r.licence_plate (1 mark)
GROUP BY c.licence_plate; (1 mark)

Total 10 marks

Question 8

A national wildlife research organisation has been using paper-based records to track animal sightings, habitats and population counts for several endangered species. **10**

Field researchers manually record this information during fieldwork and later submit the data to a central office, where it is stored in physical filing cabinets.

The organisation is considering moving to a database system to improve the efficiency of its data collection and management.

Identify FIVE (5) advantages of using a database system over the current manual system. Provide ONE (1) specific example to illustrate each advantage.

Mark scheme:

Suggested answers are not limited to:

- ***Improve data accuracy:***
  - ***Example: Field researchers can use digital forms on tablets to record animal sightings directly into the database. This reduces errors from manual data entry and ensures that information is recorded accurately and consistently.***
- ***Efficient data processing/retrieval***
  - ***Example: Researchers can quickly query the database to find all sightings of a particular species over the past year, rather than sifting through numerous paper records. This speeds up research and decision-making processes.***
- ***Enhanced Data Security***
  - ***Example: Sensitive data about endangered species' habitats can be encrypted and access-controlled in the database, ensuring that only authorised personnel can view or modify this information, protecting it from unauthorised access or loss.***
- ***Scalability***

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    - ***Example: As the organization expands its research to include more species and larger geographic areas, the database can easily scale to accommodate the increased volume of data without the need for additional physical storage space.***
- ***Data Consistency***
    - ***Example: When updating the population count of a species, the database ensures that all related records are updated simultaneously, maintaining consistency across all data points. This prevents discrepancies that might occur with manual updates in multiple locations.***
- ***Automatic calculation***
- ***Centralised data access for field researchers***
- ***Better data security***
- ***Handle large amount of data efficiently***
- ***Backup***
- ***Automated report generation***
- ***Natural disasters recovery***

1 mark for each correctly identified advantage and 1 mark for each example, up to a maximum of 10 marks.

Any suitable advantages and examples should be accepted.

Total 10 marks

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Marks

Question 9

a) A national healthcare provider has recently faced issues related to the mismanagement of their patient database. **8**

Identify FOUR (4) possible consequences of failing to properly manage their database backup and recovery processes. Provide an explanation of each consequence.

Mark scheme:

Suggested answers are not limited to:

- ***Data loss: can result in permanent loss of critical patient information.***
- ***Availability loss: service outages or system failures can make the database unavailable.***
- ***Legal and regulatory penalties: failure to backup and recovery strategies adequately can result in non-compliance with data protection regulations.***
- ***Operational disruptions: If the database cannot be quickly restored after an incident, the healthcare provider may experience significant downtime. This can halt operations, delay treatments, and impact the overall efficiency of the healthcare services.***
- ***Reputation damage: Patients trust healthcare providers to safeguard their personal and medical information. Data mismanagement can erode this trust, damage the provider's reputation, and result in a loss of patients to competitors who demonstrate better data management practices.***
- ***Increased recovery costs: When a healthcare provider does not have an effective backup and recovery strategy, the costs associated with recovering lost data can be significantly higher. This includes the expenses for emergency IT support, data recovery services, and potential overtime pay for staff working to restore systems.***

1 mark for each correctly identified area, up to a maximum of 4 marks.

1 mark for each correct description/ elaboration, up to a maximum of 4 marks.

Any other suitable answers should be accepted.

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Marks

b) A small e-commerce startup company is considering moving their database to the cloud. The company has limited technical staff and wants to focus on developing their product rather than managing IT infrastructure. **2**

Identify a suitable cloud database model for them. Provide ONE (1) reason to support your answer.

Mark scheme:

Managed (1 mark). Reduce the need for in-house database expertise (1 mark), allowing the startup to focus on product development rather than database maintenance (1 mark). Ideal for a startup with limited technical resources to handle database tasks (1 mark). Managed models offer scalability, which allows the startup to grow (1 mark).

1 mark for the correct cloud database model.

1 mark for any one of the reasons above.

Any other suitable reasons should be accepted.

Total 10 marks

Question 10

A new social media platform is developing a scalable backend to store a wide variety of data, including user profiles, real-time messages, social connections and user activity logs. The platform is considering using a NoSQL database to manage its data efficiently.

a) Define what NoSQL stands for. **1**

Mark scheme:

Non SQL/Not Only SQL/Non-relational SQL.

1 mark for any one above.

b) Identify and explain THREE (3) NoSQL data models that could be used to store different types of data for the platform. Provide ONE (1) example of data that each model would handle best. **6**

Mark scheme:

- ***Document Store (1 mark): Suitable for handling user profiles where each user may have different attributes, such as bio, interests or settings (1 mark).***

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- ***Key-Value Store (1 mark): Ideal for storing session data or caching frequently accessed information, such as user login sessions or preferences (1 mark).***
- ***Graph Database (1 mark): Best suited for managing user connections, friendships and followers, as it efficiently handles relationships (1 mark).***

1 mark for each suitable data model, up to a maximum of 3 marks.

1 mark for each correct data, up to a maximum of 3 marks.

Any other suitable answers should be accepted.

c) Consider the 3V model: Volume, Variety and Velocity and how it applies to the data management challenges the social media platform might face when handling large-scale data. **3**

Provide an explanation of each V in this context.

Mark scheme:

- ***Volume: Refers to the massive amounts of data generated by social media platforms (1 mark). This includes millions of user-generated posts, messages, likes and other interactions every day, requiring scalable storage solutions like NoSQL databases to handle the data efficiently (1 mark).***
- ***Variety: Social media platforms deal with a wide range of data types (1 mark). This includes structured data (user profiles), semi-structured data (JSON logs, user interactions), and unstructured data (images, videos), all of which must be stored and managed in a flexible way (1 mark).***
- ***Velocity: The speed at which new data is generated and processed is critical (1 mark). Social media platforms must manage real-time interactions, such as live messaging, newsfeeds, and notifications, requiring high-speed data ingestion and querying capabilities (1 mark).***

1 mark for each explanation, up to a max of 3 marks.

Any other suitable answers should be accepted.

Total 10 marks

End of Paper

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Learning Outcomes matrix

Question	Learning Outcomes assessed	Marker can differentiate between varying levels of achievement
1	1, 2	Yes
2	3,4	Yes
3	3	Yes
4	1, 4, 5	Yes
5	2, 4	Yes
6	5	Yes
7	5	Yes
8	1	Yes
9	1, 7	Yes
10	6	Yes

Grade descriptors

Learning Outcome	Fail	Referral	Pass	Merit	Distinction
Understand the concepts associated with database systems	Has little to no awareness of principles and concepts underlying theoretical frameworks and approaches and is unable to identify associated strengths or weaknesses.	Has vague awareness of principles and concepts underlying theoretical frameworks and approaches and is only superficially able to identify associated strengths or weaknesses.	Has satisfactory awareness of principles and concepts underlying theoretical frameworks and approaches and demonstrates sufficient ability to identify associated strengths and weaknesses.	Has very good awareness of principles and concepts underlying theoretical frameworks and approaches and is able in detail to identify associated strengths and weaknesses.	Has an excellent awareness of principles and concepts underlying theoretical frameworks and approaches and is comprehensively able to identify associated strengths and weaknesses.

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Understand the concepts associated with the relational model	Has little to no awareness of principles and concepts underlying theoretical frameworks and approaches and is unable to identify associated strengths or weaknesses.	Has vague awareness of principles and concepts underlying theoretical frameworks and approaches and is only superficially able to identify associated strengths or weaknesses.	Has satisfactory awareness of principles and concepts underlying theoretical frameworks and approaches and demonstrates sufficient ability to identify associated strengths and weaknesses.	Has very good awareness of principles and concepts underlying theoretical frameworks and approaches and is able in detail to identify associated strengths and weaknesses.	Has an excellent awareness of principles and concepts underlying theoretical frameworks and approaches and is comprehensively able to identify associated strengths and weaknesses.
Understand how to design and develop a database system	Demonstrates little to no ability to adequately use design principles to effectively develop an artefact to solve an identified issue.	Demonstrates an extremely limited ability to adequately use design principles to effectively develop an artefact to solve an identified issue.	Demonstrates a satisfactory ability to use design principles to effectively develop an artefact to solve an identified issue.	Demonstrates a very good ability to use design principles to effectively develop an artefact to solve an identified issue.	Demonstrates an excellent ability to use design principles to effectively develop an artefact to solve an identified issue.
Be able to develop and enhance a logical database design	Demonstrates little to no ability to adequately use design principles to effectively develop an artefact to solve an identified issue.	Demonstrates an extremely limited ability to adequately use design principles to effectively develop an artefact to solve an identified issue.	Demonstrates a satisfactory ability to use design principles to effectively develop an artefact to solve an identified issue.	Demonstrates a very good ability to use design principles to effectively develop an artefact to solve an identified issue.	Demonstrates an excellent ability to use design principles to effectively develop an artefact to solve an identified issue.

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Be able to develop and enhance a database system using SQL	Demonstrates little to no ability to adequately develop an artefact to solve an identified issue.	Demonstrates an extremely limited ability to adequately develop an artefact to solve an identified issue.	Demonstrates a satisfactory ability to develop an artefact to solve an identified issue.	Demonstrates a very good ability to develop an artefact to solve an identified issue.	Demonstrates an excellent ability to develop an artefact to solve an identified issue.
Understand the principles behind Big Data and the NoSQL databases	Displays no understanding of Big Data and NoSQL databases. Unable to recognise the key features and benefits of NoSQL over traditional databases or how Big Data impacts database design and scalability.	Shows a basic awareness of Big Data and NoSQL databases. Recognises some key aspects but fails to understand the broader implications and technical considerations involved.	Demonstrates satisfactory comprehension of Big Data principles and the operational basics of NoSQL databases. Can distinguish NoSQL databases from traditional databases and understands their use cases.	Exhibits a strong grasp of the principles behind Big Data and NoSQL databases. Can articulate the advantages of NoSQL databases in handling large datasets and different data types.	Has an in-depth understanding of Big Data and NoSQL databases. Can critically evaluate and recommend NoSQL database solutions for complex Big Data problems, demonstrating a sophisticated understanding of scalability and performance optimisation.
Understand the concepts and principles of Cloud database	Lacks understanding of the fundamental concepts and principles of Cloud databases. Cannot identify the main advantages or the operational model of Cloud databases.	Has a superficial grasp of Cloud databases. Recognises some concepts but cannot articulate how Cloud databases differ from on-premises solutions or their benefits.	Has a solid understanding of Cloud database concepts and principles. Understands the benefits of scalability, availability, and managed services provided by Cloud databases.	Possesses a very good understanding of Cloud databases. Can discuss the importance of Cloud databases in modern architectures and understands their significance in providing flexible and scalable data storage solutions.	Demonstrates an excellent and comprehensive understanding of the key concepts and principles of Cloud databases. Can describe in detail how Cloud databases offer advantages in terms of accessibility, maintenance, and scalability, and can suggest appropriate use cases for Cloud databases.

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